



OPERATING SYSTEM DESIGN

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PROCESS & THREAD MANAGEMENT

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- Fork system call is used for creating a new process
- The new process is the child process which work concurrently with the parent process
- After a new child process is created, both processes will execute the next instruction following the fork() system call.
- It takes no parameters and returns an integer value
- Fork is called once but return twice

Negative Value: creation of a child process was unsuccessful.

Zero: Returned to the newly created child process.

Positive value: Returned to parent or caller. The value contains process ID of newly created child process.

```
#include <stdio.h>
#include <unistd.h>
int main()
{
    fork();

    printf("Hello\n");
    return 0;
}
```

Output:

Hello

Hello

```
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid;

    // Fork a child process
    pid = fork();

    if (pid < 0) {
        fprintf(stderr, "Fork failed\n");
        return 1;
    }
}
```

```
else if (pid == 0) {  
    // Child process  
    printf("Child process is running (PID: %d)\n", getpid());  
} else {  
    // Parent process  
    printf("Parent process is running (PID: %d)\n", getpid());  
}  
  
return 0;  
}
```

```
#include <stdio.h>
#include <pthread.h>
void *thread_function(void *arg) {
    printf("Thread is running\n");
    return NULL;
}
int main() {
    pthread_t tid;

    // Create a thread
    if (pthread_create(&tid, NULL, thread_function, NULL) !=
0) {
        fprintf(stderr, "Thread creation failed\n");
        return 1;
    }
}
```



```
// Wait for the thread to finish
if (pthread_join(tid, NULL) != 0) {
    fprintf(stderr, "Thread join failed\n");
    return 1;
}

printf("Main program is running\n");

return 0;
}
```



THANK YOU

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